

Chemical lumbar sympathectomy in the treatment of recalcitrant erythromelalgia



Wen-Hui Wang, MD,^a Long Zhang, MD,^{b,c} Guo-Xiang Dong, MD,^b Ting-Ting Sun, BS,^a Zhi-Miao Lin, MD,^d Yong Yang, MD,^d and Xuan Li, MD,^b *Beijing, China*

ABSTRACT

Objective: Erythromelalgia is highly disabling and treatment is often very challenging. There have been solitary case reports that it might benefit from sympathectomy. This study sought to evaluate the short-term and long-term efficacy of chemical lumbar sympathectomy (CLS) for treatment of recalcitrant erythromelalgia and try to identify a CLS-responsive subset.

Methods: Patients with recalcitrant erythromelalgia were recruited from a tertiary hospital over a 10-year period. L3 to L4 CLS was performed using 5% phenol. The pain intensity score (visual analog scale [VAS] 0-10) was assessed before CLS and at 1 day, 1 week, 3 months, 6 months, 1 year, and 2 years after CLS. A VAS decrease of 90%-100% is defined as complete response, 60%-89% as major partial response. Relapse was defined by a return of a VAS score of 5 or higher. *SCN9A* gene mutations were screened.

Results: Thirteen patients were enrolled, with a median age of 15 years. The mean follow-up was 6.2 ± 3.8 years. *SCN9A* gene mutation was identified in five patients having family histories. The VAS was 8.2 ± 2.0 at baseline; it decreased to 4.9 ± 2.7 at 1 day and 1.9 ± 3.0 at 1 week after CLS. Nine patients (69.2%) achieved complete response at 1 week after CLS, including three patients with *SCN9A* gene mutation. Among the three complete response patients having the gene mutation, two reverted to major partial response and one relapsed at 2 years after CLS. Among the six complete response patients without mutation, five maintained complete response and one relapsed. Among the four patients who did not achieve complete response, one patient died at 3.5 months and one patient had an amputation performed at 4 months after CLS.

Conclusions: CLS provides a valid option for the treatment of recalcitrant erythromelalgia. It takes about 1 week to achieve full efficacy. Relapse may occur, especially in patients with an *SCN9A* gene mutation. (J Vasc Surg 2018;68:1897-905.)

Keywords: Chemical lumbar sympathectomy; Erythromelalgia; Gene mutation; *SCN9A*

Erythromelalgia (also called erythermalgia in some of the literature) is a rare syndrome characterized by a burning sensation of the extremities associated with erythema as well as increased skin temperature. The feet and legs are typically involved; warmth intensifies the symptoms and relief may be achieved by cooling. In severe cases, symptoms may be continuous instead of intermittent. Immersion feet and/or legs with maceration, infection, ulceration, and gangrene may set in, setting up a vicious cycle of unbearable pain. In a

follow-up study of 168 erythromelalgia patients from the Mayo clinic, 45 patients died a mean of 6.3 years after the diagnosis, including three from suicide.¹

Treatment of erythromelalgia is often very challenging. A great variety of therapeutic options have been reported, including but not limited to, (1) anti-inflammatory drugs, such as aspirin, indomethacin, and piroxicam; (2) immunosuppressants, such as corticosteroids, cyclosporine, and plasma exchange; (3) beta-adrenergic blocking agents including propranolol; (4) alpha-adrenergic blocking agents including phenoxybenzamine; (5) anticonvulsant drugs, such as gabapentin, clonazepam, and carbamazepine; (6) opioids; (7) tricyclic antidepressants; (8) adrenergic drugs, including adrenaline, ephedrine, and isoproterenol; (9) coronary vasodilators such as nitroglycerin; (10) vasodilators including nitroprusside; (11) antiserotonin drugs, including methysergide, cyproheptadine, and pizotifen; (12) serotonin receptor agonists such as sumatriptan; (13) traditional Chinese medicine; and (14) physical methods, including biofeedback and epidural blocks. However, none of these therapeutic approaches have provided consistent or long-term relief. Recalcitrant erythromelalgia often results, despite multiple treatments.

Sympathectomy has been a treatment choice to improve peripheral blood flow. It also interrupts sympathetic–nociceptive coupling by a direct neurolytic

From the Department of Dermatology,^a Department of Interventional Radiology and Vascular Surgery,^b and Wound Healing Center,^c Peking University Third Hospital; and the Department of Dermatology, Peking University First Hospital.^d

Funded by grants from Beijing Higher Education Young Elite Teacher Project (Grant No: YETP0072 to W.W.) and Peking University Third Hospital Scientific Research Foundation for the Returned Overseas Scholars (Grant No: 77434-01 to L.Z.).

Author conflict of interest: none.

Correspondence: Long Zhang, MD, Department of Interventional Radiology and Vascular Surgery, Peking University Third Hospital, 49 North Garden Road, Haidian District, Beijing 100191, China (e-mail: longzh2000@126.com).

The editors and reviewers of this article have no relevant financial relationships to disclose per the JVS policy that requires reviewers to decline review of any manuscript for which they may have a conflict of interest.

0741-5214

Copyright © 2018 by the Society for Vascular Surgery. Published by Elsevier Inc. <https://doi.org/10.1016/j.jvs.2018.05.226>

action on nociceptive fibers that may, theoretically, relieve the pain of erythromelalgia. There have been 11 cases reported worldwide in which sympathectomy had been performed on erythromelalgia patients; nine responded,²⁻⁹ but two patients from one pedigree carrying the F1449V *SCN9A* gene mutation did not experience pain relief after sympathectomy.^{10,11} Sympathectomy can be performed by several means: open surgically, telescopically, chemically through the injection of phenol or anhydrous ethanol under radiographic guidance, or by radiofrequency. Of these methods, chemical lumbar sympathectomy (CLS) does not require general anesthesia; in addition, it is economically attractive, less invasive, and can be performed repeatedly. CLS has been used in our hospital for more than 20 years, mainly for the treatment of ischemic diseases, Raynaud disease, and palmoplantar hyperhidrosis, and recently for livedo reticularis, and has been shown to be a safe treatment.¹²⁻¹⁵ To explore both the short-term and long-term efficacy and safety of CLS in the treatment of recalcitrant erythromelalgia, and try to identify a CLS-responsive subset, we performed a prospective study over a 10-year period. The study was approved by the local ethics committee and all patients signed written informed consent.

METHODS

Study population. Patients with recalcitrant erythromelalgia fulfilling the following criteria were recruited: (1) a convincing history of redness, heat, and painful extremities; (2) involvement of the lower extremities; (3) at least five medications have been used and the pain intensity score (visual analog scale [VAS] 0-10) remains greater than or equal to 5; and (4) experiencing a wait of at least 1 to 2 weeks for behavioral education (trying not to overuse ice, cold water, and fans) and wound care, to minimize the escalating effect of immersion feet. Exclusion criteria were an autoimmune history or autoimmune laboratory abnormalities and upper extremity, ear, neck, or torso involvements.

A wide range of laboratory tests was performed including but not limited to a complete blood count, urinalysis, biochemistry tests, rheumatism tests, immunoglobulins, immunofixation electrophoresis, chest radiographs, electrocardiogram, and abdominal B-ultrasound examination. All patients were recruited from dermatology and/or interventional radiology and vascular surgery clinics in our hospital between 2005 and 2015.

CLS. CLS on ganglia L3 and L4 was performed under fluoroscopy guidance with detailed technical points described in our previous report.¹⁵ Briefly, the patient was in the lateral decubitus position with a pad on the flank opposite to the operation side and the lower limbs were kept flexed. The skin surface and the entire needle

ARTICLE HIGHLIGHTS

- **Type of Research:** Prospective, single-center, cohort study
- **Key Findings:** Chemical lumbar sympathectomy in 13 patients with recalcitrant erythromelalgia resulted in a complete response in nine patients at 1 week, although recurrences were more common in patients with *SCN9A* gene mutation.
- **Take Home Message:** This study suggests that chemical lumbar sympathectomy is a valid option to treat recalcitrant erythromelalgia.

tracts were anesthetized with 0.5% lidocaine without vasoconstrictors. The needle was inserted at approximately 7 to 8 cm from the dorsomedian line to facilitate outflank introduction according to body shape. The first needle was inserted at L3 level, then L4 level. The needle was introduced toward the medial portion of the vertebral body and its tip extended to approximately the one-third vertical dividing line of the vertebral body, which was behind the anterior fascia of the psoas major muscle. Positioning was confirmed by the injection of iodated contrast (Fig 1). After aspiration of the syringe to ensure that no puncture accidents would occur, 2 to 3 mL of 5% aqueous solution of phenol was injected at each puncture site. The other side was performed in the same manner. The average skin temperature in the shin, plantar, and dorsum of the foot was recorded. The temperature was measured using an infrared thermometer with accuracy of 0.1°C. Satisfactory sympathetic interruption was defined as an increase in the average temperature of at least 2°C at 5 minutes after CLS.

Efficacy and safety assessment. A pain intensity score (VAS 0-10) was used. Patients and caregivers were trained for how to score the VAS by the same doctor before enrolling in the study. A final score was agreed by consensus of the patients and caregivers. The efficacy variable was graded as 1 (complete response; VAS decrease of 90%-100%); 2 (major partial response; VAS decrease of 60%-89%); 3 (minor partial response; VAS decrease of 30%-59%); or 4 (no response; VAS decrease of ≤29%). Relapse was defined by a return of VAS 5 or greater during after at least 2 years of follow-up. Safety assessments included the monitoring and recording of all adverse events and tolerability to treatment.

***SCN9A* gene analysis.** Genomic DNA samples (isolated from the peripheral blood of the patients and their relatives) were subjected to polymerase chain reaction amplification. Twenty-eight pairs of primers were designed to amplify the 26 exons and their flanking sequences. The polymerase chain reaction products

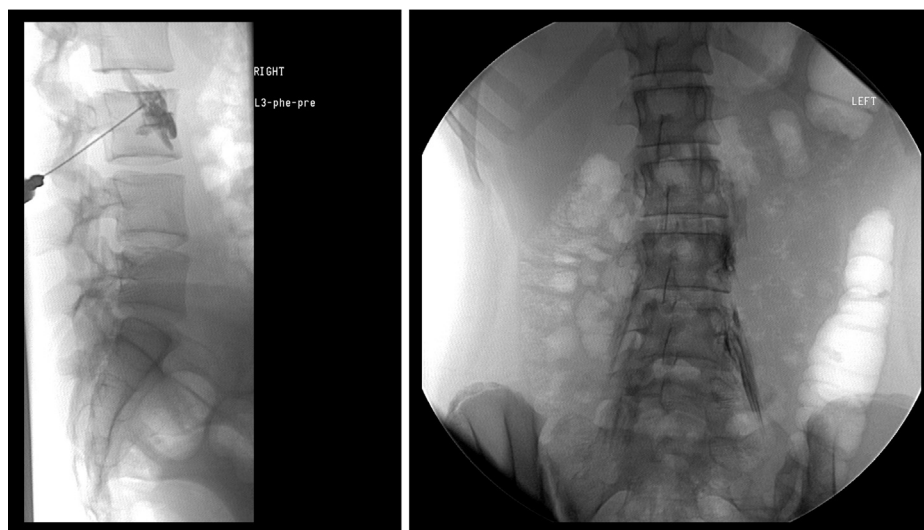


Fig 1. The introduction and positioning of the needles in chemical lumbar sympathectomy (CLS). **Left,** Lateral radiograph. **Right,** Anteroposterior radiograph. Note the injection of the drug was behind anterior fascia with the target at gray rami communicans, rather than in the sympathetic trunk.

were subjected to gel electrophoresis purification and direct sequencing using an ABI Prism 377 sequencer (Applied Biosystems, Foster City, Calif).

RESULTS

Thirteen patients were enrolled in this study (Table); nine female and four male, with a median age of 15 years (range, 11-52 years). Ten patients were younger than 18 years of age. The median disease duration was 36 months (range, 2-120 months). Patients had attempted a mean of 12.3 ± 7.8 adjuvant medications for their pain with inadequate pain control before CLS was performed. Ulcers were seen in nine patients (Fig 2). The mean follow-up time was 6.2 ± 3.8 years.

Among the 13 patients, 5 patients had a family history of erythromelalgia and the *SCN9A* gene mutation was found in all of these five families: c.4894 G→A (A1632T), c. 2573T→A (L858H), c.1185C→G (N395K), c.3947T→C (V1316A), and c.2477T→A (F826Y) (Patients 6, 7, 9, 10, and 12; Fig 3). No *SCN9A* mutations were detected in the other eight patients; these patients also had negative family histories of erythromelalgia.

The mean VAS value of the thirteen patients before CLS was 8.2 ± 2.0 . The mean VAS values 1 day, 1 week, and 3 months after CLS treatment were 4.9 ± 2.7 , 1.9 ± 3.0 , and 1.9 ± 2.9 , respectively (Fig 4). One patient died at 3.5 months (patient 13) and one patient required amputation at 4 months (patient 12) after CLS. Among the nine patients with ulcers, seven had healed at 3 months after CLS treatment.

At 1 week after CLS, nine patients achieved complete response. VAS dynamics through 2 years for these nine patients are also shown in Fig 4. All nine patients maintained complete response at 6 months after CLS. Four patients maintained complete response for 6.2 to

12.2 years after CLS. One patient (patient 9, with N395K *SCN9A* gene mutation) experienced relapse (VAS of 8) at 1 year after CLS treatment; ulcers recurred and enlarged, and the patient self-treated with corticosteroid (about 30 mg/d) intermittently. A status of partial efficacy resulted, yet the efficacy was gradually lost and the pain progressed to a VAS of 9 at 2 years after CLS. Repeated CLS was performed, resulting in no response and the patient died 4.7 years after the first CLS treatment. A second patient (patient 8, without an *SCN9A* gene mutation) experienced relapse (VAS of 5) at 2 years after CLS. Repeated CLS treatment was undertaken, and a status of complete response was achieved again. However, this status was only maintained for 2 months; VAS reached a level of 5 at 6 months after the second CLS. A third CLS was performed; however, the patient did not respond to the third treatment. The patient then took ergot caffeine, resulting in minor partial response. The other three patients who did not maintain complete response at 2 years after CLS were assessed at a VAS of 2 or 3 (patient 5, without an *SCN9A* gene mutation; patients 6 and 7, with *SCN9A* gene mutation) and the pain could be tolerated with any treatment.

At 1 week after CLS, one patient achieved major partial response (patient 10, with a V1316A *SCN9A* gene mutation). This patient relapsed (VAS of 8) at 2 years after CLS and an additional CLS was not performed. Sumatriptan (25 mg/d) initially provided some relief (VAS of 3); however, 1 year later, sumatriptan gradually lost effect, and the VAS returned to a level of 8. In addition, ulcers recurred and enlarged. A combination treatment of gabapentin (600 mg/d), carbamazepine (400 mg/d), and sumatriptan (25 mg/d) maintained VAS levels at 5 to 6 and the ulcers persisted.

Table. Disease characteristics and chemical lumbar sympathectomy (CLS) efficacy

Case	Gender	Age, years	Disease duration, months	Subacute onset ^a	Precipitant	Comorbidity
Patients who achieved complete response at 1 week after CLS						
1	Female	15	3	Yes	None	Congenital spina bifida and hexadactylism
2	Female	18	108	Chronic	None	None
3	Male	15	4	Yes	Developed erythromelalgia 1 month after trauma (fracture of transverse process of left L1-L3 and right L5 in a traffic accident)	None
4	Male	16	3	Yes	None	None
5	Female	22	120	Chronic	None	Popliteal artery entrapment syndrome
6 ^b	Female	14	36	Chronic	None	None
7 ^b	Female	11	36	Chronic	None	Persistent hypokalemia with renin, angiotensin, aldosterone elevation
8	Male	52	24	Chronic	Developed erythromelalgia 7 years after bone marrow transplantation	Myelodysplastic syndromes-M2 Leukemia; smoking
9 ^b	Male	11	48	Chronic	None	Phenylketonuria with normal physical and mental states
Patients who achieved major partial response at 1 week after CLS						
10 ^b	Female	14	120	Chronic	None	Hypertension
Patients who achieved minor partial response at 1 week after CLS						
11	Female	15	2	Chronic	None	Physical and mental retardation, micrognathia
12 ^b	Female	14	96	Chronic	None	Hypertension, Hypothyroidism
Patients who got no response at 1 week after CLS						
13	Female	13	96	Chronic	None	None

PLT, Platelets; VAS, visual analog scale.

^aA subacute temporal profile, defined as disease crescendo to peak intensity in <21 days.¹⁶^bPatient 6, 7, 9, 10, and 12 had positive family histories, SCN9A gene mutation was detected (Fig 3).

Table. Continued.

Steroid responsiveness	Aspirin responsiveness	PLT ($\times 10^9/L$)	VAS before CLS	VAS 1 week after CLS	VAS 2 years after CLS	Follow-up, years
No response	No response	300-400	10	0	0	12.2
Minor partial response	Minor partial response	>400	5	0	0	6.3
No response	No response	300-400	10	0	0	6.2
Not used	No response	Normal	8	0	0	6.2
Not used	No response	Normal	6	0	2	12.0
No response	No response	300-400	6	0	2	6.0
No response	No response	>400	10	1	3	11.1
No response	No response	Normal	5	0	5	6.8
Major to minor partial response	No response	Normal	8	0	9	4.7
Minor partial response	No response	>400	8	3	8	5.4
Not used	No response	Normal	10	6	0	2.0
No response	No response	>400	10	7	Amputation (at 4 months after CLS)	1.0
No response	No response	>400	10	8	Death (at 3.5 month after CLS)	0.3



Fig 2. Clinical pictures of the 13 patients with recalcitrant erythromelalgia. At 1 week after chemical lumbar sympathectomy (CLS), patients 1-9 achieved complete response, patient 10 achieved major partial response, patients 11 and 12 achieved minor partial response, and patient 13 got no response. At 2 years after CLS, patient 8-10 relapsed.

Three patients (23.1%) reported mild pain localized to the thigh area with no need for analgesic relief, and the pain resolved spontaneously within 12 hours in two patients and 3 days in one patient. No other adverse events were observed.

DISCUSSION

The largest case series on erythromelalgia reported so far have included 168 cases studied between 1970 and 1994 in the Mayo clinic. However, ulcers were only seen in 10 patients (6%).¹ In this current study, we performed CLS on 13 patients with recalcitrant erythromelalgia, including nine with ulcers and five with gene mutations. Given the rare nature of erythromelalgia and the fact that only recalcitrant cases were recruited, our study population is sizeable. There was a female predominance (9:4) in our recalcitrant erythromelalgia cohort; this predominance has also been described in the Mayo Clinic (122:46)¹ and in Olmsted County, Minnesota (28:5),¹⁷ for total erythromelalgia. All of our patients were evaluated before CLS and at 1 day, 1 week, 3 months, 6 months, 1 year, and 2 years after CLS treatment unless death or amputation occurred, with a mean follow-up of 6.2 years. Thus, in this study, the dynamic short-term and long-term efficacy rates were followed.

Erythromelalgia is a devastating disease and treatment is very challenging. In one report, treatment with immunosuppressants in the form of oral corticosteroids or

plasma exchange was reported to be the highest rate of a very helpful response¹; however, this rate was only achieved in 33.3% of patients,¹ or 29% in another report.¹⁶ Beside the low response rate, patients usually need to take drug lifelong and the side effects can be fatal. In our current study, we found a 69.2% rate (9 of 13) of achieving a complete response in recalcitrant erythromelalgia with CLS, which is a much higher rate than that achieved with oral corticosteroids; moreover, the procedure showed a tolerable safety profile. Even in patients with *SCN9A* gene mutation, the rate of achieving complete response at 1 week reached 60.0% (3 of 5). The phenol used in CLS is a permanent sclerosant agent, so the effect should be long-lasting, as is seen in most of our patients. However, some patients experienced a relapse, especially patients with the *SCN9A* gene mutation. We hypothesize that it may be attributable to regeneration or compensation of residual nerve fibers, a proposition meriting further study.

CLS is the result of direct obliteration of sympathetic nerves; its action onset is assumed to be immediate. In our experience, it usually takes less than 10 minutes to achieve full efficacy in the treatment of ischemic disease, Raynaud disease, paronychia, hyperhidrosis, and livedo reticularis. However, we found it took about 1 week to achieve full efficacy for the treatment of erythromelalgia, although most patients received obvious relief within 1 day. This phenomenon suggests that the recovery of

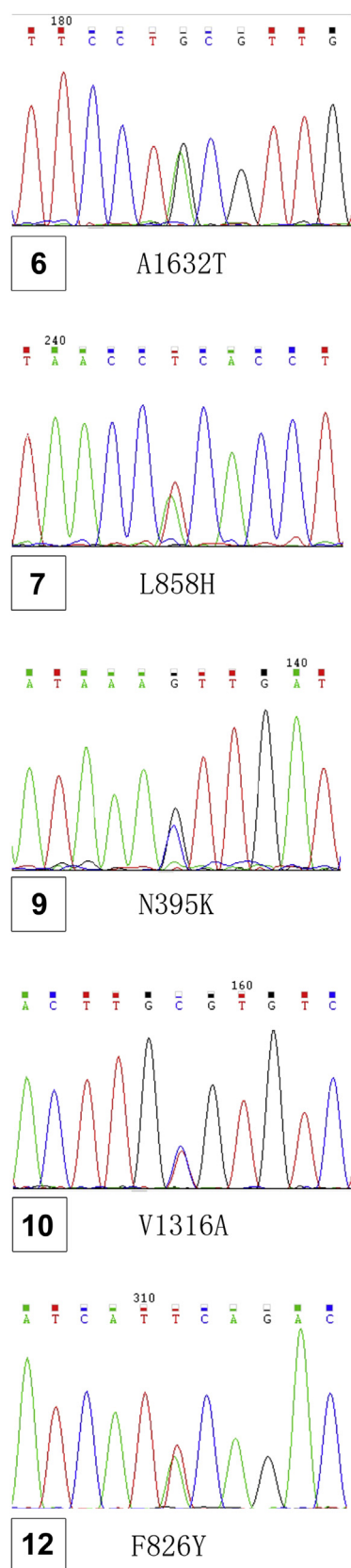


Fig 3. Mutation analysis of SCN9A gene. An SCN9A gene mutation was found in the 5 patients who had positive family histories (patients 6, 7, 9, 10, and 12).

abnormal sympathetic–nociceptive coupling in erythromelalgia is a gradual process. Another consideration in pain relief may be that the initial pain is concurrent with wound pain.

Although CLS is easily performed, exquisite precision is imperative. Our experience in performing CLS for more than 20 years—not limited to the treatment of erythromelalgia in this study—is that injection of the drug behind the anterior fascia with the target at gray rami communicans, rather than sympathetic trunk, may achieve comparable sympathetic interruption and be much safer. The major complications of inadvertent damage to adjacent structures, particularly ureteropelvic, could be largely avoided by this way. The key technique points of CLS were illustrated in detail in our previous report of CLS in idiopathic livedo reticularis.¹⁵ The most common adverse event in CLS is genitofemoral neuralgia localized to the thigh and groin, as seen in 23.1% of our patients, which was mild and resolved spontaneously within 12 hours to 3 days. Contraindications to CLS are deformity of the spine, severe bleeding tendency or thrombolytic therapy within 2 weeks, any kind of infection in the disinfection area and needle path, unstable vital signs, pregnancy, or any condition that causes the patient to be unable to comply with treatment procedures.

For the phenotype-responsiveness correlations, it has been reported that in a 31-patient cohort, a traumatic, surgical, or infectious erythromelalgia precipitant and subacute temporal profile to disease zenith (<21 days) may predict corticosteroid responsiveness.¹⁶ High-dose corticosteroids (≥ 40 mg prednisone daily over ≥ 5 days, or ≥ 1 g/day methylprednisolone for 5 days) was significantly associated with corticosteroid responsiveness.¹⁶ Thus, a transient golden window has been proposed, where useful corticosteroid intervention may exist before irreversible nociceptive remodeling and central sensitization occurs.¹⁶ In the current study, only one patient seemed to fit the golden window model (patient 3: traumatic precipitant, subacute profile to disease zenith); however, this patient was unresponsive to corticosteroids at a high dose. In our study, 10 patients had undergone corticosteroid treatment, six of them at high dose levels. However, only one patient (patient 9, with an SCN9A gene mutation, no disease precipitant, chronic profile to disease zenith) had achieved major partial response to corticosteroid, and the corticosteroid effectiveness gradually diminished, and the patient died.

A special type of erythromelalgia associated with thrombocytopenia had been described as an aspirin-responsive type. In this type, the burning pain and erythema are usually distributed unilaterally or asymmetrically, with preferential involvement of one or more toes, forefoot soles, or fingertips, and the platelet count is always elevated ($>400.00 \times 10^9/L$).^{18,19} In our study,

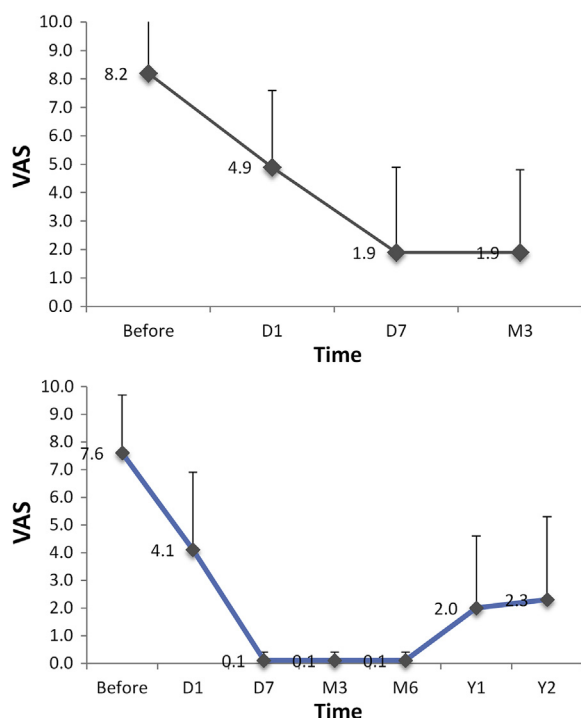


Fig 4. The pain intensity score on a visual analog scale (VAS) before and after chemical lumbar sympathectomy (CLS). **Top**, VAS dynamics through 3 months in all of the 13 patients. **Bottom**, VAS dynamics through 2 years in 9 patients who achieved a complete response at 1 week after CLS.

5 of 13 patients had a persistent platelet count of higher than $400.00 \times 10^9/L$ (three with an *SCN9A* gene mutation and two without); only one patient (without a gene mutation) showed a minor partial response to aspirin.

In the analyses of genotype-responsiveness correlations, a Canadian family with a V400M *SCN9A* gene mutation responded favorably to carbamazepine,²⁰ an English family with an I848T *SCN9A* gene mutation responded favorably to combination of gabapentin and carbamazepine,²¹ and a Chinese patient with a V872G *SCN9A* gene mutation responded favorably to mexiletine.²² In our study, the five patients with the *SCN9A* gene mutation did not experience favorable response to these drugs. Our experience in dealing with patients with an *SCN9A* gene mutation was that some patients responded favorably to sumatriptan initially, but the drug gradually lost effect with time.

Another aim of this study was to identify a CLS-responsive subset of patients. Although certain patients with an *SCN9A* gene mutation responded to CLS initially, efficacy was not maintained in these patients, suggesting a less beneficial role for CLS in patients with the *SCN9A* gene mutation. No predilection of gender, age, disease duration, subacute or chronic onset, and precipitant events were found between CLS responsive and nonresponsive patients.

CONCLUSIONS

This study indicates that CLS is a valid treatment option for recalcitrant erythromelalgia, especially in patients without the *SCN9A* gene mutation. It takes about 1 week to achieve full efficacy, which is a relative longer period than for other diseases. Relapse may occur, especially in patients with the *SCN9A* gene mutation; in such cases, repeat CLS treatment is of limited benefit.

We thank the patients and their families, without whom this study would not have been possible.

AUTHOR CONTRIBUTIONS

Conception and design: WW, LZ, GD, ZL, YY, XL

Analysis and interpretation: WW, LZ, GD, TS, ZL, YY, XL

Data collection: WW, LZ, TS

Writing the article: WW, LZ

Critical revision of the article: WW, LZ, GD, TS, ZL, YY, XL

Final approval of the article: WW, LZ, GD, TS, ZL, YY, XL

Statistical analysis: WW, LZ

Obtained funding: WW, LZ

Overall responsibility: LZ

REFERENCES

1. Davis MD, O'Fallon WM, Rogers RS, Rooke TW. Natural history of erythromelalgia: presentation and outcome in 168 patients. *Arch Dermatol* 2000;136:330-6.
2. Zhang L, Wang WH, Li LF, Dong GX, Zhao J, Luan JY, et al. Long-term remission of primary erythromelalgia with R1150W polymorphism in *SCN9A* after chemical lumbar sympathectomy. *Eur J Dermatol* 2010;20:763-7.
3. Cerci FB, Kapural L, Yosipovitch G. Intractable erythromelalgia of the lower extremities successfully treated with lumbar sympathetic block. *J Am Acad Dermatol* 2013;69:e270-2.
4. Bang YJ, Yeo JS, Kim SO, Park YH. Sympathetic block for treating primary erythromelalgia. *Korean J Pain* 2010;23:55-9.
5. Lee JY, Sim WS, Kang RA, Lee EK, Yang JY, Kim DY. Lumbar sympathetic pulsed radiofrequency treatment for primary erythromelalgia: a case report. *Pediatr Dermatol* 2017;34:e47-50.
6. Seishima M, Kanoh H, Izumi T, Niwa M, Matsuzaki Y, Takasu A, et al. A refractory case of secondary erythromelalgia successfully treated with lumbar sympathetic ganglion block. *Br J Dermatol* 2000;143:868-72.
7. Shiga T, Sakamoto A, Koizumi K, Ogawa R. Endoscopic thoracic sympathectomy for primary erythromelalgia in the upper extremities. *Anesth Analg* 1999;88:865-6.
8. Nakajima Y, Koizumi K, Hirata T, Hirai K, Sakamoto A, Shimizu K. Successful thoracoscopic sympathectomy for primary erythromelalgia in the upper extremities. *J Thorac Cardiovasc Surg* 2004;52:524-6.
9. Li X, Tu YR, Lin M, Lai FC. Sympathectomy for primary erythromelalgia. *China Modern Doctor* 2007;45:49-50.
10. Dib-Hajj SD, Rush AM, Cummins TR, Hisama FM, Novella S, Tyrrell L, et al. Gain-of-function mutation in Nav1.7 in familial erythromelalgia induces bursting of sensory neurons. *Brain* 2005;128:1847-54.
11. Finley WH, Lindsey JR, Fine JD, Dixon GA, Burbank MK. Autosomal dominant erythromelalgia. *Am J Med Genet* 1992;42:310-5.
12. Dong GX, Zhao J. Chemical sympathectomy for ischemia of lower extremity. *J Beijing Med University* 1998;1:59-60.

13. Dong GX, Zhao J, Luan JY. Chemical lumbar sympathectomy for the treatment of lower limb ischemia. *Chin J Gen Surg* 2002;1:30-2.
14. Dong GX. Chemical lumbar sympathectomy. *Chin J Prac Surg* 2004;4:245-6.
15. Wang WH, Zhang L, Li X, Zhao J, Zhuang JM, Dong GX. Chemical lumbar sympathectomy in the treatment of idiopathic livedo reticularis. *J Vasc Surg* 2015;62:1018-22.
16. Pagani-Estévez GL, Sandroni P, Davis MD, Watson JC. Erythromelalgia: identification of a corticosteroid-responsive subset. *J Am Acad Dermatol* 2017;76:506-11.
17. Reed KB, Davis MD. Incidence of erythromelalgia: a population-based study in Olmsted County, Minnesota. *J Eur Acad Dermatol Venereol* 2009;23:13-5.
18. Michiels JJ, Drenth JP, van Genderen PJ. Classification and diagnosis of erythromelalgia and erythermalgia. *Int J Dermatol* 1995;34:97-100.
19. Drenth JP, Michiels JJ. Erythromelalgia and erythermalgia: diagnostic differentiation. *Int J Dermatol* 1994;33:393-7.
20. Fischer TZ, Gilmore ES, Estacion M, Eastman E, Taylor S, Melanson M, et al. A novel Nav1.7 mutation producing carbamazepine-responsive erythromelalgia. *Ann Neurol* 2009;65:733-41.
21. Natkunarajah J, Atherton D, Elmslie F, Mansour S, Mortimer P. Treatment with carbamazepine and gabapentin of a patient with primary erythermalgia (erythromelalgia) identified to have a mutation in the SCN9A gene, encoding a voltage-gated sodium channel. *Clin Exp Dermatol* 2009;34:e640-2.
22. Choi JS, Zhang L, Dib-Hajj SD, Han C, Tyrrell L, Lin Z, et al. Mexiletine-responsive erythromelalgia due to a new Na(v)1.7 mutation showing use-dependent current fall-off. *Exp Neurol* 2009;216:383-9.

Submitted Mar 10, 2018; accepted May 24, 2018.